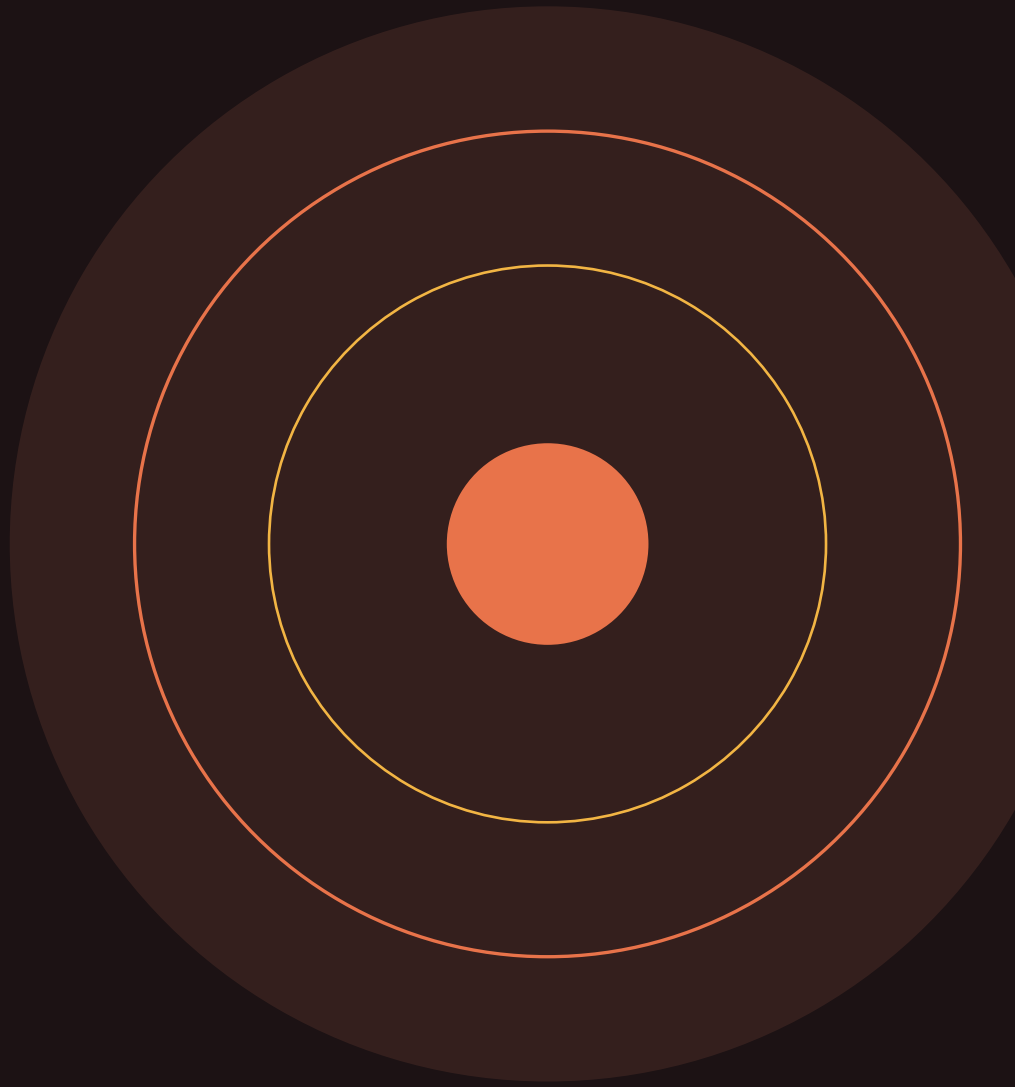
BDM CAPSTONE PROJECT · 16 JULY 2026

Utilizing Sales Pattern Analytics to Customize Marketing and Optimize Stock at a Pharmacy



Six months of one Bengaluru chemist's own billing data — 25,512 transaction lines — turned into four validated analyses.

Abhishek Ramesh Kumar
IITM Online BS Degree Program · Bengaluru



[Pharmacy]

Chemist & druggist · Bengaluru

BUSINESS BACKGROUND

A chemist sitting on unread data

MODEL B2C retail chemist & druggist — prescription medicines, OTC, supplements, personal care.

SYSTEM Daily billing captured in simple invoicing software. Until this project, no analysis was run on it.

STOCKING Reorder decisions made on experience and supplier recommendation, not transaction evidence.

MARKETING One uniform SMS to every customer. No segmentation, no targeted promotion.

SCOPE 25,512 lines · 13,355 invoices · 820 customers · 40 products · Jan–Jun 2024

WHY THIS PROJECT

Problem Statement & Objectives

03

The billing system records every sale. Nothing was done with it — no way to tell which customers were worth retaining, which products sold together, or which items carried the greatest stock-out risk.

1

Purchase trends & repeat behaviour

Determine customer purchase trends and repeat-purchasing behaviour.

2

Targeted marketing

Enable customized promotions through customer categorisation — currently absent.

3

Inventory planning

Address lost cross-selling and stock imbalance from missing trend and association analysis.

OBJECTIVES DELIVERED

- Segment customers by Recency, Frequency & Monetary value
- Mine product associations via market-basket analysis
- Classify all 40 products by revenue and demand stability
- Forecast revenue and demand, validated on a hold-out month
- Translate every finding into a numbers-based recommendation

DATA FOUNDATION

What the Data Actually Contains

04

₹91.95L
Total revenue, 6 months

13,355
Unique invoices

820
Unique customers

40
Products, 14 categories

FIELDS CAPTURED · 18 COLUMNS

Invoice / Customer / Product IDs	Identifier	Invoice_No, Customer_ID, Batch_No
Category, Brand, Payment mode	Categorical	Category, Brand, Payment_Mode
Quantity, Price, Revenue	Numerical	Quantity, Unit_Price, GST, Total_Amount
Dates & times	Date / Time	Invoice_Date, Invoice_Time, Expiry_Date

INVOICE VALUE PROFILE

Mean	**₹688.53**
Median	**₹395.08**
Coeff. of variation	**135.07%**
Avg items / invoice	**1.91**

A CV of 135% means basket value is highly uneven — the average invoice is not a typical one.

All 25,512 rows complete across 18 columns — no missing values, no duplicates. Academic-use authorisation from [Pharmacy] .

METHODOLOGY

How the Analysis Was Built

05

PREPROCESSING

- Checked missing values, duplicates, invoice-level consistency
- Converted invoice dates to date-time; grouped by month and day
- Reconstructed invoice-level product baskets from invoice numbers
- Aggregated customer-level Recency / Frequency / Monetary values
- Built daily and monthly demand series per product

01

RFM Segmentation

Recency, Frequency, Monetary quartile scoring → 4 customer segments

02

Market Basket Analysis

Apriori on multi-item invoices → support, confidence, lift

03

ABC-XYZ Classification

Revenue tier (ABC) crossed with demand-variability tier (XYZ)

04

Demand Forecasting

Linear trend, validated by train / hold-out split (MAE, MAPE)

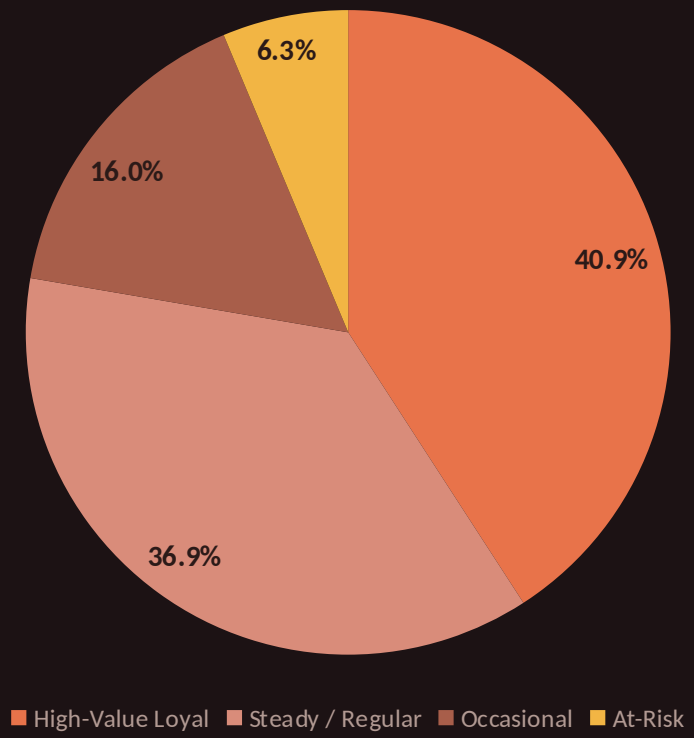
TOOLS Python — Pandas, NumPy, mlxtend (Apriori), Matplotlib Microsoft Excel — source workbook, live formulas, native charts

CUSTOMER ANALYTICS

Who Actually Pays the Bills

06

Share of ₹91.95L revenue



CUSTOMERS PER SEGMENT · CIRCLE SIZED BY COUNT



High-Value Loyal	Steady / Regular	Occasional	At-Risk
26% of base	38% of base	24% of base	13% of base

The At-Risk circle is the smallest group and the smallest revenue slice — but it is the only one still winnable.

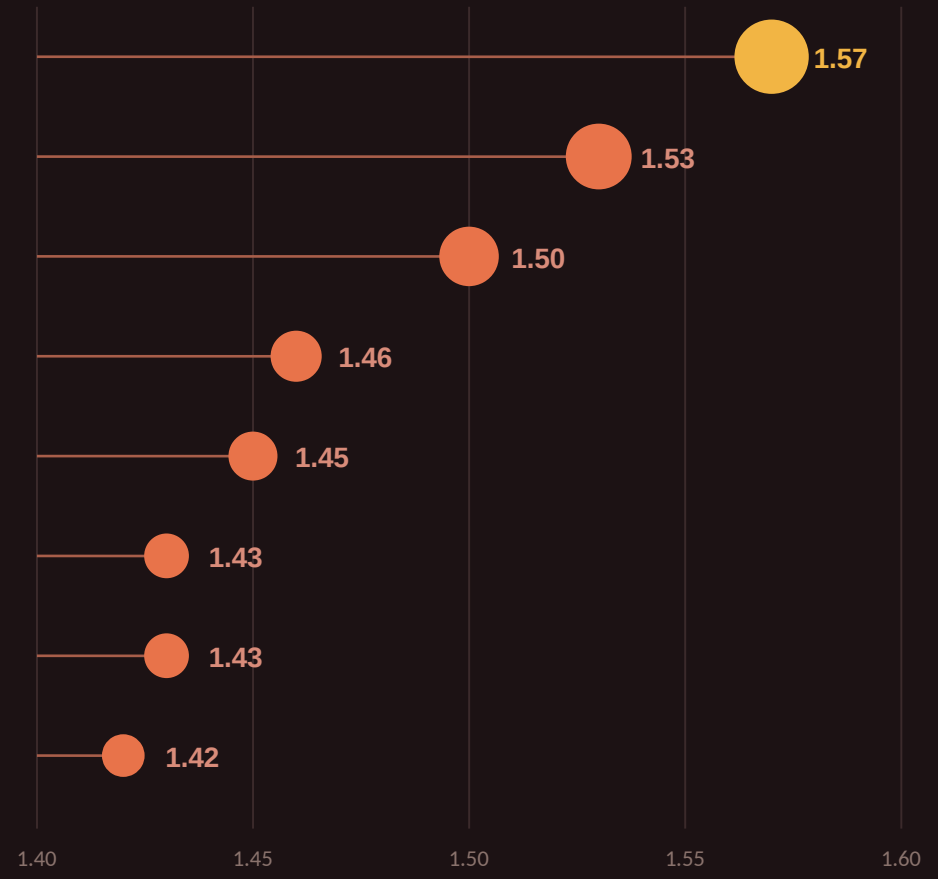
64% of customers hold 78% of revenue. The 104 At-Risk customers are only 6.31% of revenue today — but they are identifiable by name, and addressable before they lapse.

CROSS-SELL OPPORTUNITY

What Gets Bought Together

07

PRODUCT PAIR	LIFT (1.40 → 1.60)
Clonazepam 0.5 + Amlodipine 5	1.57
Alprazolam 0.25 + Amlodipine 5	1.53
Clonazepam 0.5 + Alprazolam 0.25	1.50
Clonazepam 0.5 + Glimepiride 2	1.46
Atorvastatin 10 + Metoprolol 50	1.45
Levothyroxine 50 + Atorvastatin 10	1.43
Glimepiride 2 + Metformin 500	1.43
Atorvastatin 10 + Glimepiride 2	1.42



55.40%

of invoices (7,399 of 13,355) contain 2 or more products

22.29%

of revenue comes from the 8 products in these pairs

1.91 → 2.2

target basket size once these pairs are actively bundled

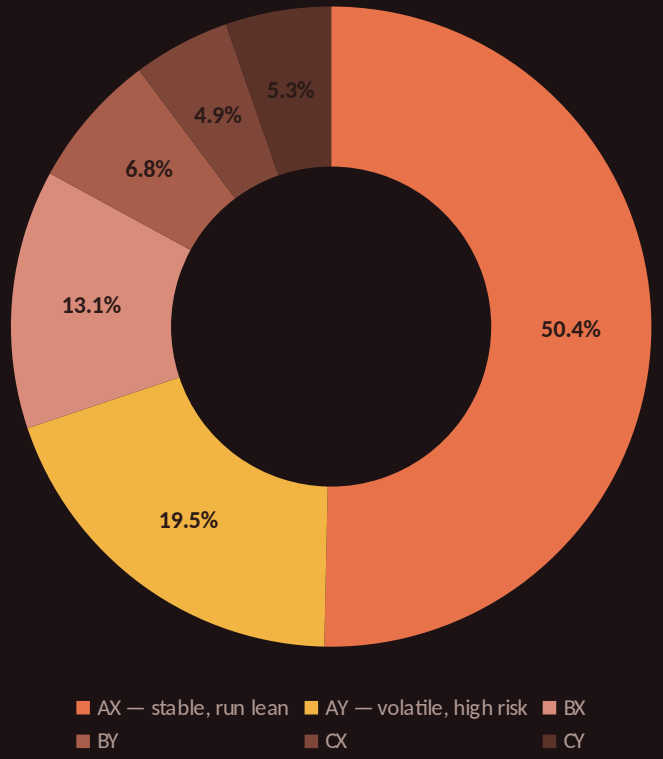
Every strong link sits among chronic-therapy medicines — sleep/anxiety, cardiac, diabetes, thyroid — refilled on a predictable cycle.

INVENTORY PRIORITY

Where the Stock-Out Risk Sits

08

ABC-XYZ: share of revenue by class



5 AY products — 19.51% of revenue riding on unstable demand

High revenue, high variability. These carry the store's greatest stock-out exposure, so each gets an explicit safety stock and reorder point rather than a judgement call.

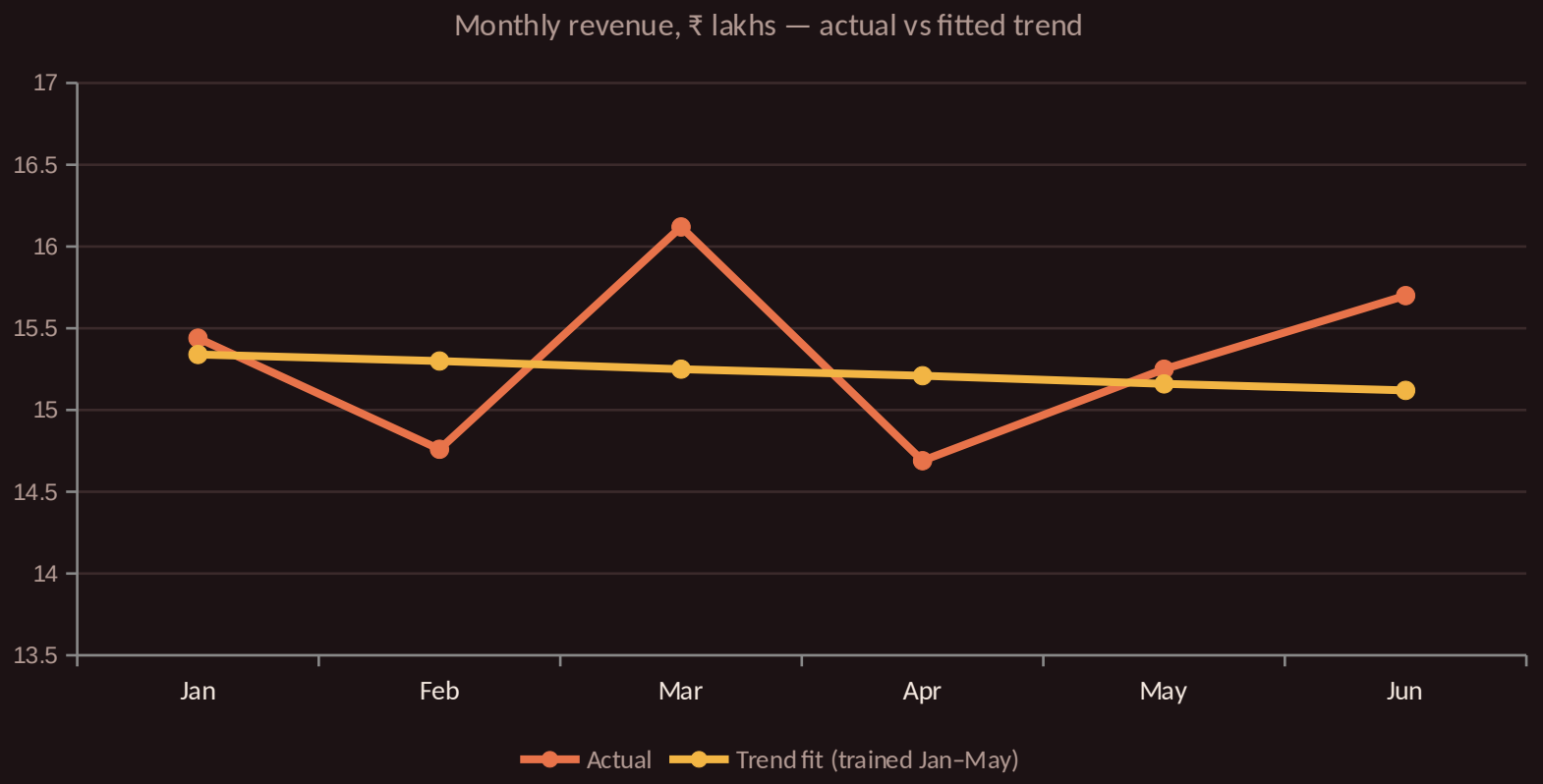
PRODUCT	DEMAND CV	SAFETY	REORDER
Salbutamol Inhaler	32.65%	19 units	**43 units**
Montelukast 10mg	26.97%	16 units	**38 units**
Volini Spray 100g	23.48%	19 units	**41 units**
Calcium + D3	21.42%	16 units	**38 units**
Vitamin D3 60K	21.35%	15 units	**34 units**

15 of 40 products (Class A) drive 69.85% of revenue. Ten are AX — stable enough to run lean. The other five are the ones to watch.

FORWARD PLANNING

Can the Trend Be Trusted?

09



3.73%

Hold-out MAPE — trained Jan–May, tested against actual June

₹15.46L

Projected July 2024 revenue, from the full 6-month trend

RELIABLE Diabetes, Eye/Ear, Neuro-Sleep (MAPE < 10%)

PRELIMINARY Vitamins, Respiratory, Skin Care (MAPE > 55%) — six months is too short to confirm seasonality here

FROM ANALYSIS TO ACTION

Recommendations

FINDING	RECOMMENDED ACTION
Loyal + Steady segments hold 78% of revenue	**Loyalty recognition — priority billing, small refill discounts**
104 At-Risk customers, 6.31% of revenue, lapsing	**SMS / WhatsApp win-back and refill reminders, by name**
8 named product pairs, lift 1.42–1.57	**Bundle refill reminders; place the pairs adjacent at the counter**
5 AY products: high revenue, unstable demand	**Apply the safety-stock and reorder points from slide 08**
Forecast reliability varies sharply by category	**Use the trend directly only for Diabetes / Eye-Ear / Neuro-Sleep**
Findings live across separate analyses today	**Build one monthly-refreshed dashboard (Power BI or Excel)**

CONCLUSION

From instinct to evidence

- Six months of the store's own transaction data — 25,512 lines, 13,355 invoices — became four completed, validated analyses rather than descriptive summaries.
- 213 high-value customers, 8 named product pairs and 5 highest-risk products are now individually identifiable, each with a concrete target attached.
- Together these resolve all three problem statements from the original proposal: purchase trends, targeted marketing, inventory planning.

4
analyses completed

213
high-value customers

8
cross-sell pairs

3.73%
forecast MAPE

Thank you